a. Iodine

Iodine compound are broad spectrum and considered effective for a variety of bacteria, mycobacteria, fungi and viruses. The alcoholic tincture of iodine is highly active against gram positive organisms and so is used as a skin antiseptic. It stains the skin. Iodine combines with microbial protein and inhibits their function.

b. Chloride

Chloride is also used as a gas to maintain a low microbial count in drinking water. Chlorine together with ammonia called chloramines are used to sanifige glasswall and eating ulenits. Sodium hypochlorite (NaOCl) is one of the most widely used chlorine containing disinfectants. Low concentrations (2-500ppm) are active against vegetative bacteria, fungi and most viruses. Rapid sporicidal action can be obtained around 2500ppm, however this concentration is very corrosive so should be limited in its use. High concentrations are also irritating to the mucous membranes, eyes and skin. Chlorine compounds are rapidly inactivated by light and some metals so fresh solutions should always be used. Hypochlorites should never be mixed with acids or ammonia as this will result in the release of toxic chlorine gas.

c. Iodophores

They combinations of iodine and organic molecules are called Iodophores. They include wescodine, betadine and previdone. These iodophore contains surface active agents. They cause less irritation to the skin than free Iodine and do not stain. They are used for cleaning wounds and as a general purpose laboratory disinfectant for discarded jars.

4. Heavy Metals

For many years the ions of heavy metals such as mercury, silver, arsenic, zinc, and copper were used as germicides. More recently these have been superseded by other less toxic and more effective germicides. Many heavy metals are more bacteriostatic than bactericidal. There are a few exceptions. 1% solution of Silver nitrate is often applied to the eyes of infants to prevent ophthalmic gonorrhea. Silver sulfadiazine is used on burns. Copper sulfate is an effective algicide used in lake and swimming pools to retard the growth of algae.



In many hospitals, Erythromycin is used instead of Silver nitrate because it is effective against *Chlamydia* as well as *Neisseria*.

Heavy metals combine with sulfhydryl (SH) groups of proteins and inactivate them. High concentration of metallic salts, particularly those of mercury, silver and copper coagulate cellular proteins that results in damage or death of the microbial cell. The most toxic heavy metals are the mercury, silver, and copper.

5. Quaternary Ammonium Compounds (Quats)

The most widely used surface active agents are the cationic detergents, especially the quaternary ammonium compounds (quats).

Quaternary Ammonium compounds are strongly bactericidal against Gram positive bacteria and less active against gram negative bacteria. These include

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